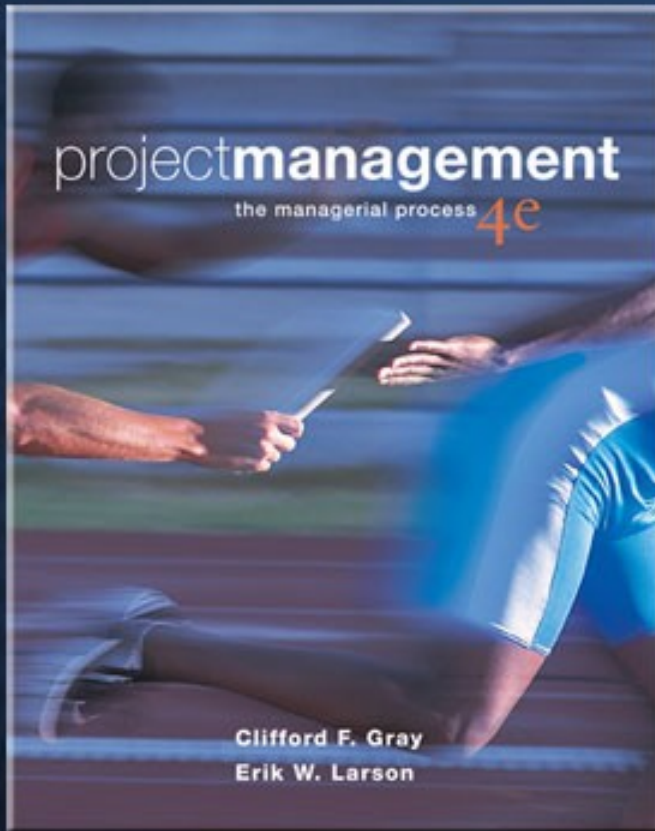
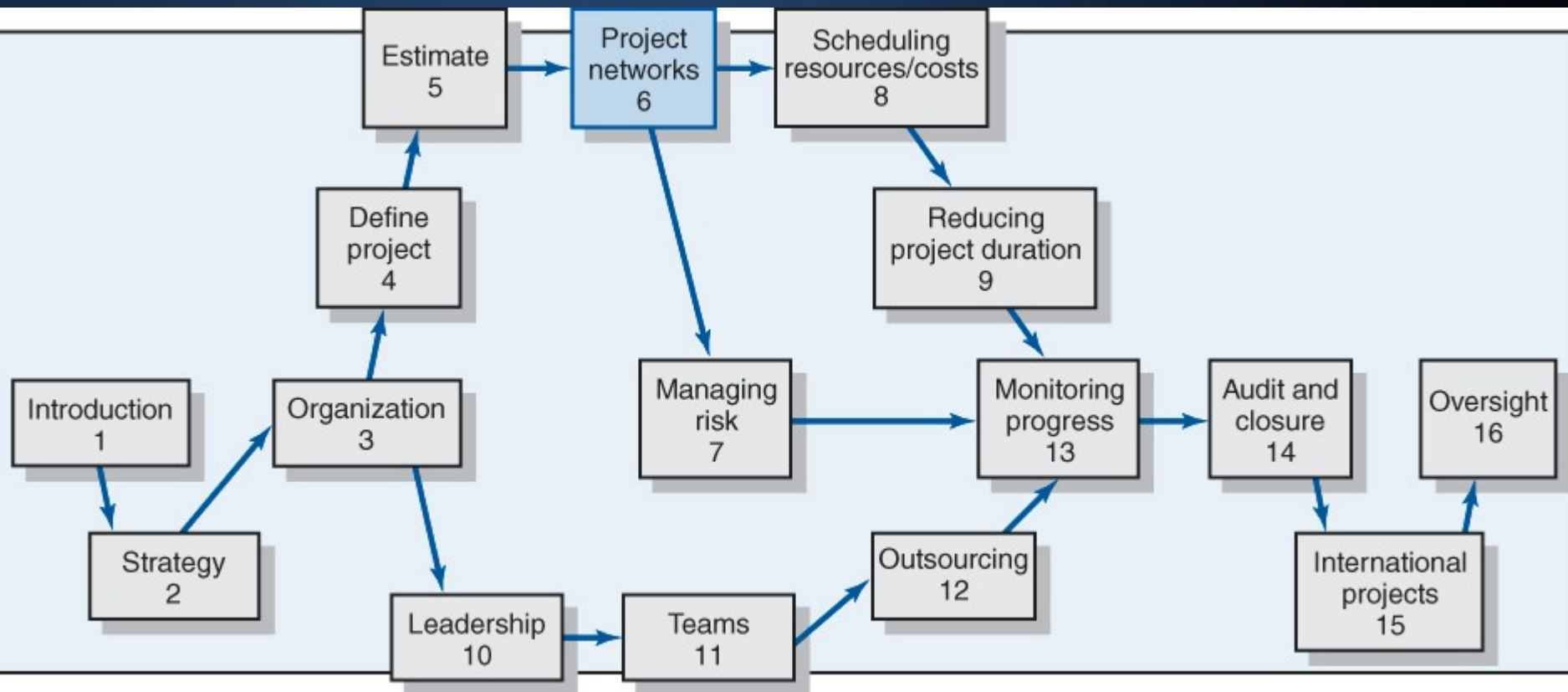




Developing a Project Plan



Developing the Project Plan

■ The Project Network

- A flow chart that graphically depicts the sequence, interdependencies, and start and finish times of the project job plan of activities that is the ***critical path*** through the network
 - Provides the basis for scheduling labor and equipment
 - Provides an estimate of the project's duration
 - Provides a basis for budgeting cash flow
 - Highlights activities that are “critical” and should not be delayed
 - Help managers get and stay on plan

From Work Package to Network

WBS/Work Packages to Network

Lowest element	Circuit board	
 Organization Units	Design cost account	<u>Design</u> WP D-1-1 Specifications WP D-1-2 Documentation
	Production cost account	<u>Production</u> WP P-10-1 Proto 1 WP P-10-2 Final Proto 2
	Test cost account	<u>Test systems</u> WPT-13-1 Test
	Software cost account	<u>Software</u> WP S-22-1 Software preliminary WP S-22-2 Software final version

FIGURE 6.1

From Work Package to Network (cont'd)

WBS/Work Packages to Network (cont'd)

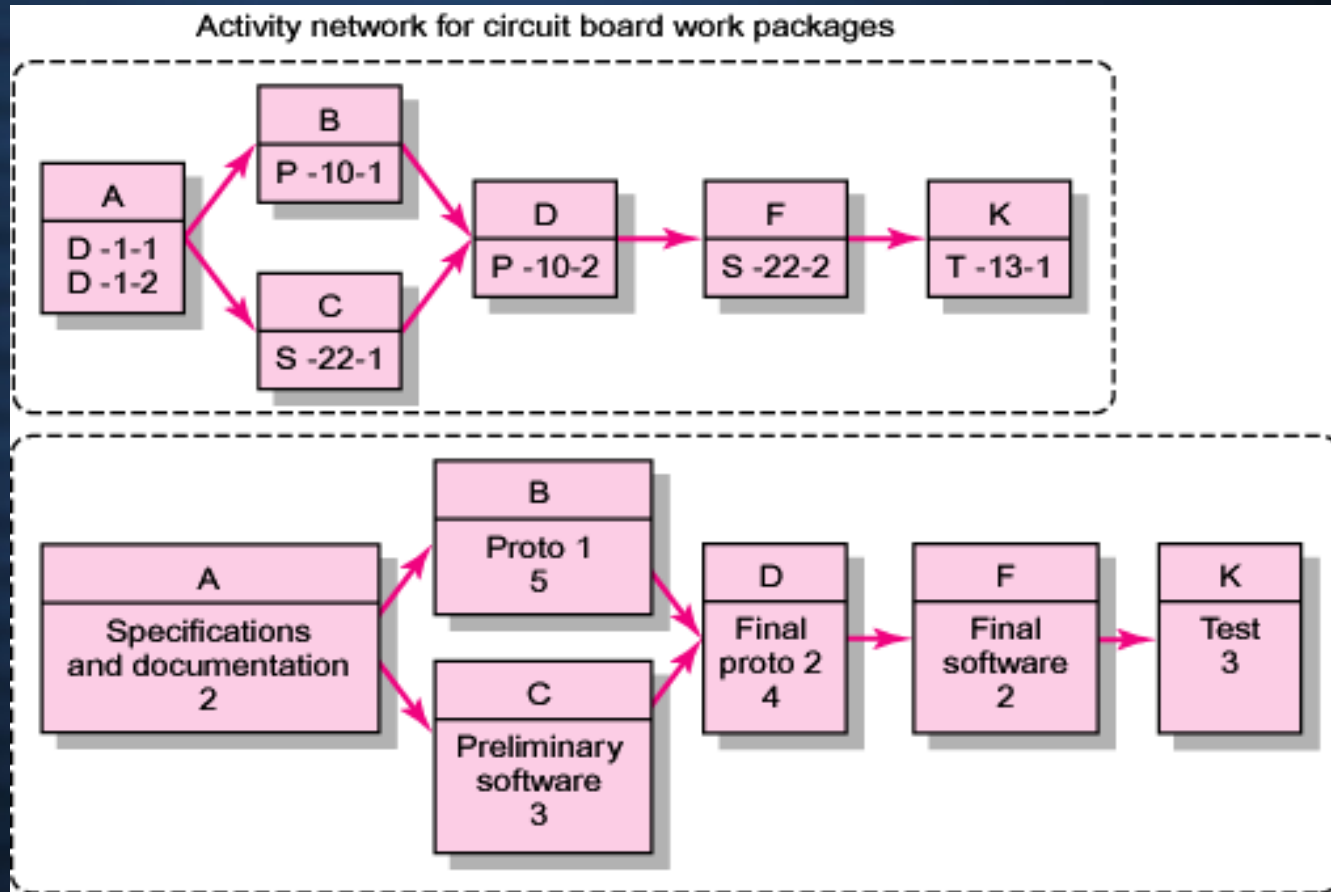
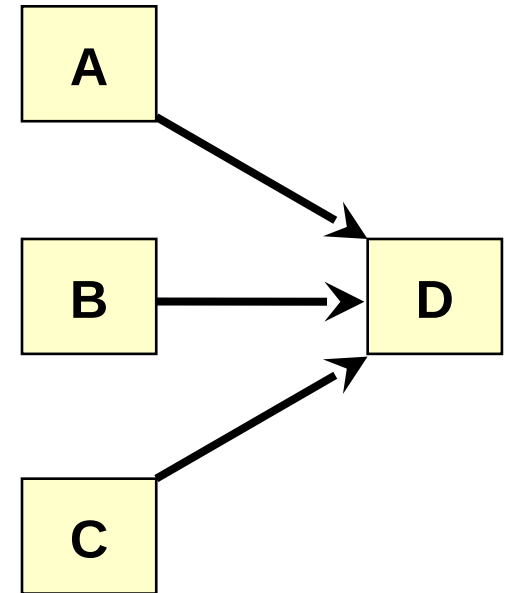


FIGURE 6.1 (cont'd)

Constructing a Project Network

■ Terminology

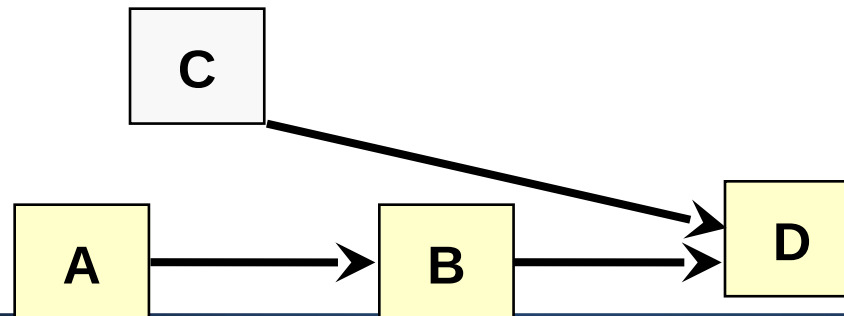
- ❑ **Activity:** an element of the project that requires time.
- ❑ **Merge activity:** an activity that has two or more preceding activities on which it depends.
- ❑ **Parallel (concurrent) activities:** Activities that can occur independently and, if desired, not at the same time.



Constructing a Project Network (cont'd)

■ Terminology

- **Path:** a sequence of connected, dependent activities.
- **Critical path:** the longest path through the activity network that allows for the completion of all project-related activities;
the shortest expected time in which the entire project can be completed. Delays on the critical path will delay completion of the entire project.



(Assumes that minimum of A + B > minimum of C in length of times to complete activities.)

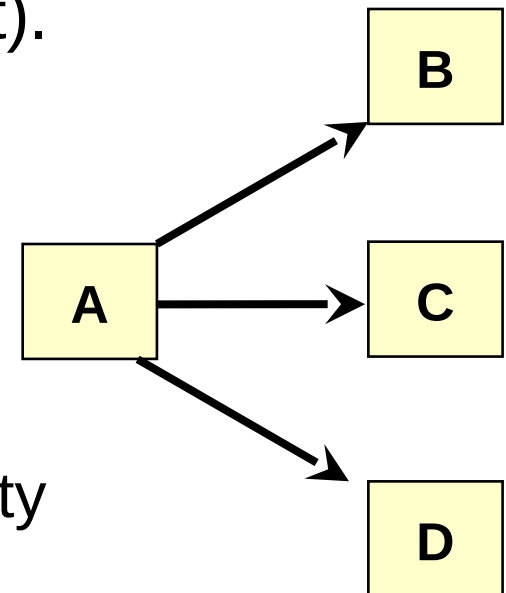
Constructing a Project Network (cont'd)

■ Terminology

- **Event:** a point in time when an activity is started or completed. It does not consume time.
- **Burst activity:** an activity that has more than one activity immediately following it (more than one dependency arrow flowing from it).

■ Two Approaches

- Activity-on-Node (AON)
 - Uses a node to depict an activity
- Activity-on-Arrow (AOA)
 - Uses an arrow to depict an activity



Basic Rules to Follow in Developing Project Networks

- Networks typically flow from left to right.
- An activity cannot begin until all of its activities are complete.
- Arrows indicate precedence and flow and can cross over each other.
- Identify each activity with a unique number; this number must be greater than its predecessors.
- Looping is not allowed.
- Conditional statements are not allowed.
- Use common start and stop nodes.

Activity-on-Node Fundamentals

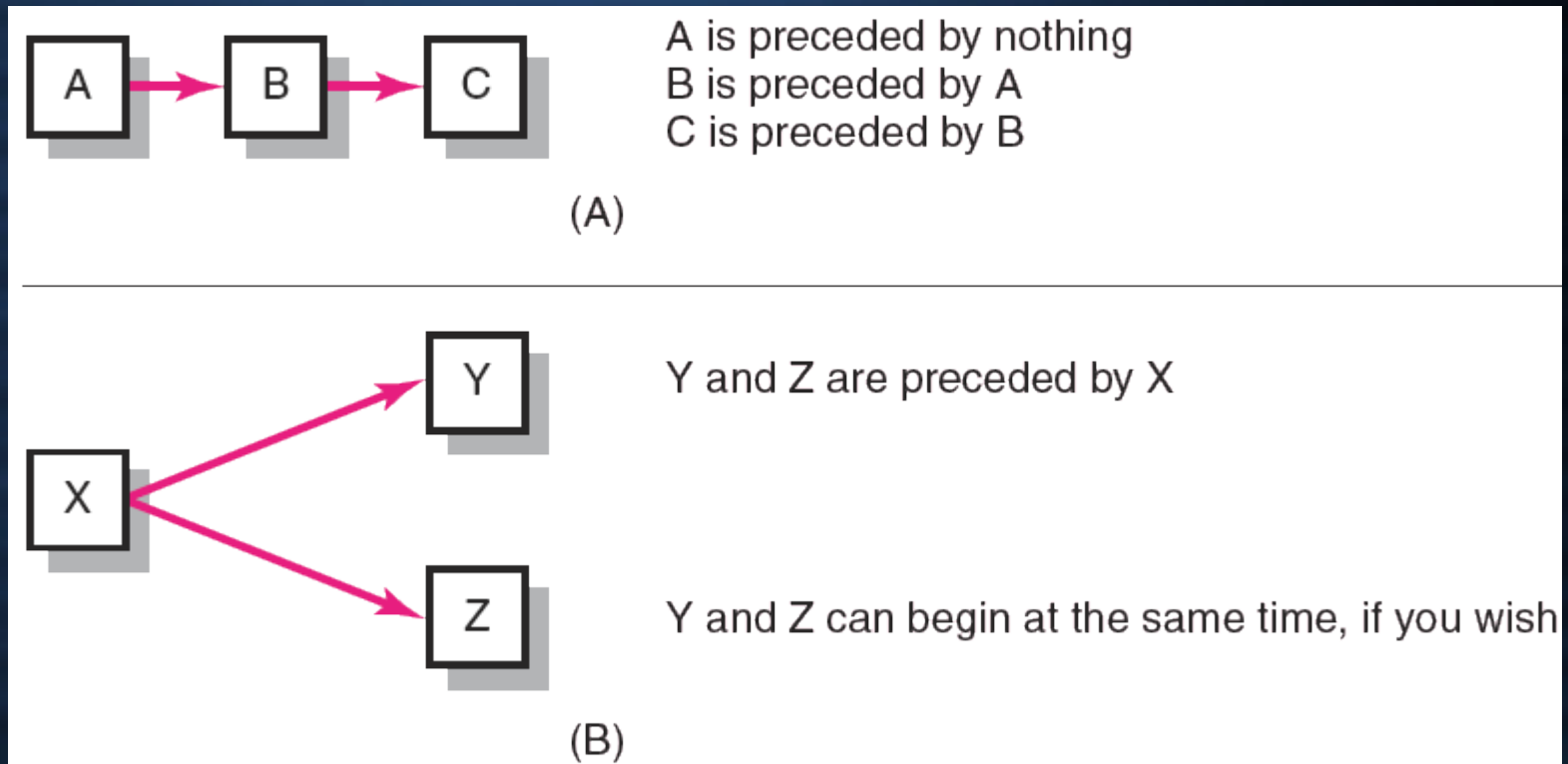
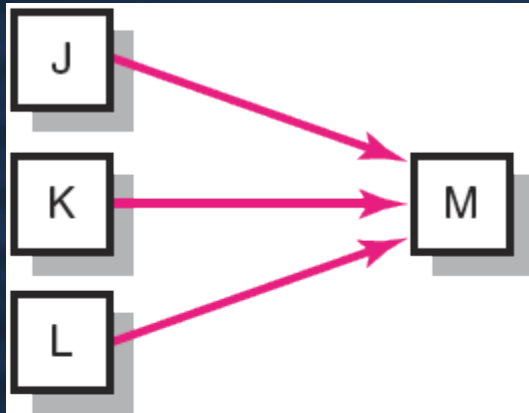


FIGURE 6.2

Activity-on-Arrow Fundamentals (cont'd)

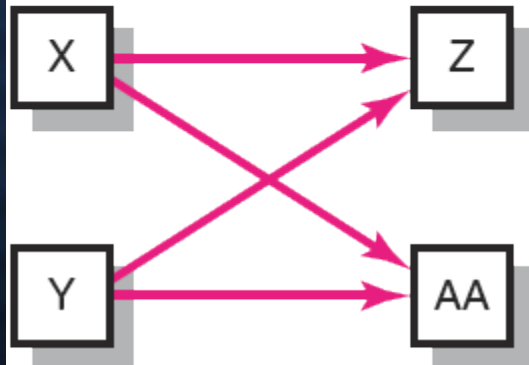


J, K, & L can all begin at the same time, if you wish (they need not occur simultaneously)

but

All (J, K, L) must be completed before M can begin

(C)



Z is preceded by X and Y

AA is preceded by X and Y

(D)

FIGURE 6.2 (cont'd)

Network Information

KOLL BUSINESS CENTER County Engineers Design Department

Activity	Description	Preceding Activity
A	Application approval	None
B	Construction plans	A
C	Traffic study	A
D	Service availability check	A
E	Staff report	B, C
F	Commission approval	B, C, D
G	Wait for construction	F
H	Occupancy	E, G

TABLE 6.1

Koll Business Center—Partial Network

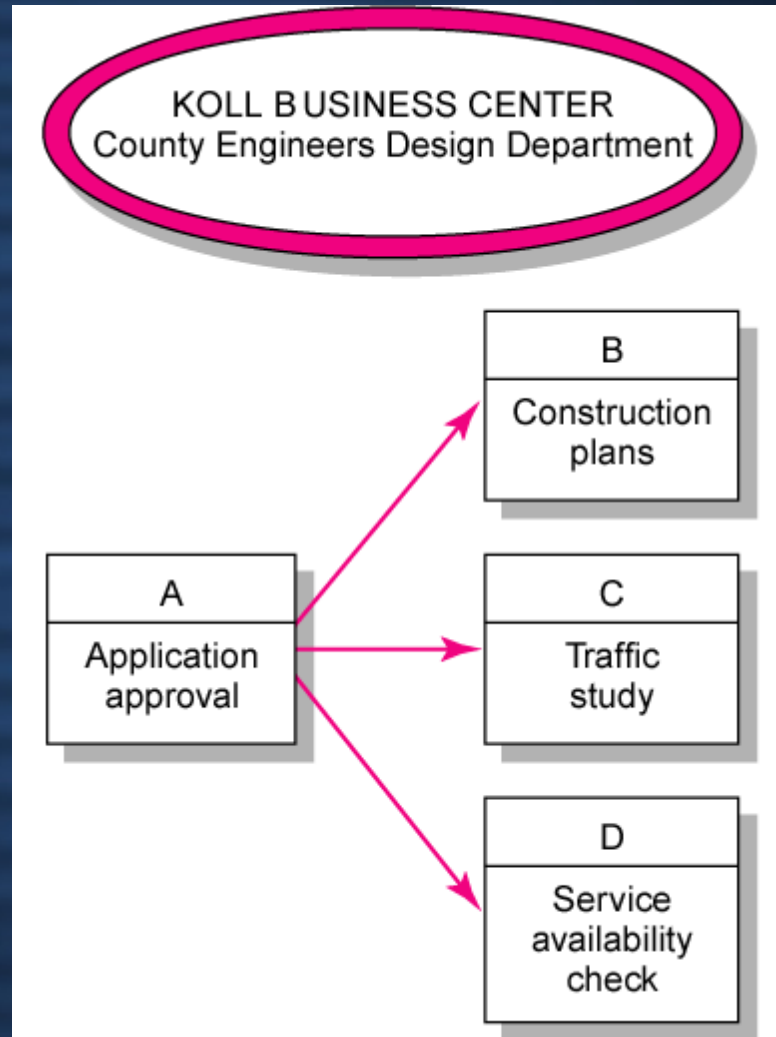


FIGURE 6.3

Koll Business Center—Complete Network

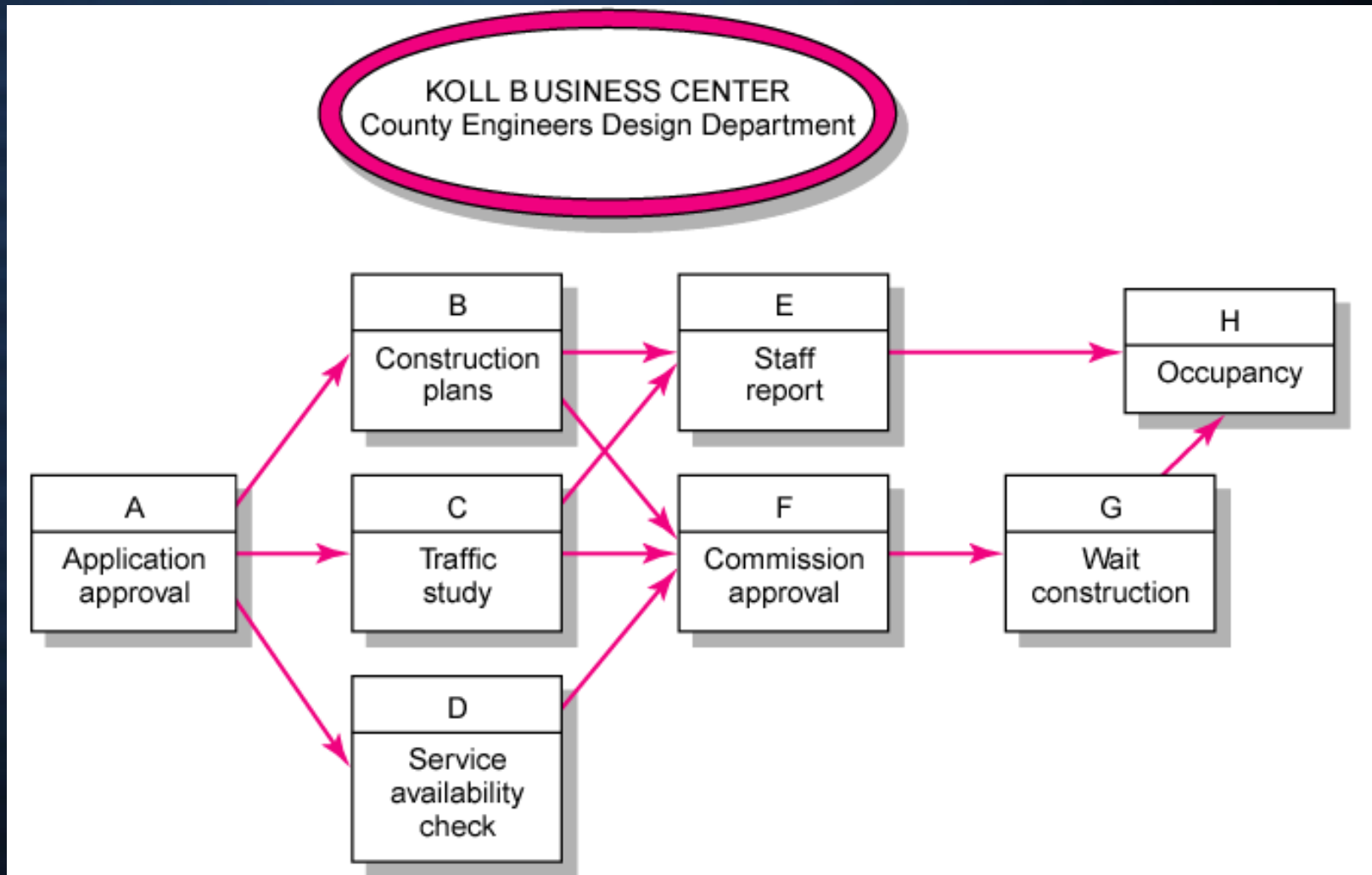


FIGURE 6.4

Network Computation Process

- Forward Pass—Earliest Times
 - How soon can the activity start? (early start—ES)
 - How soon can the activity finish? (early finish—EF)
 - How soon can the project finish? (expected time—ET)
- Backward Pass—Latest Times
 - How late can the activity start? (late start—LS)
 - How late can the activity finish? (late finish—LF)
 - Which activities represent the critical path?
 - How long can it be delayed? (slack or float—SL)

Network Information

KOLL BUSINESS CENTER County Engineers Design Department

Activity	Description	Preceding Activity	Activity Time
A	Application approval	None	5
B	Construction plans	A	15
C	Traffic study	A	10
D	Service availability check	A	5
E	Staff report	B, C	15
F	Commission approval	B, C, D	10
G	Wait for construction	F	170
H	Occupancy	E, G	35

TABLE 6.2

Activity-on-Node Network

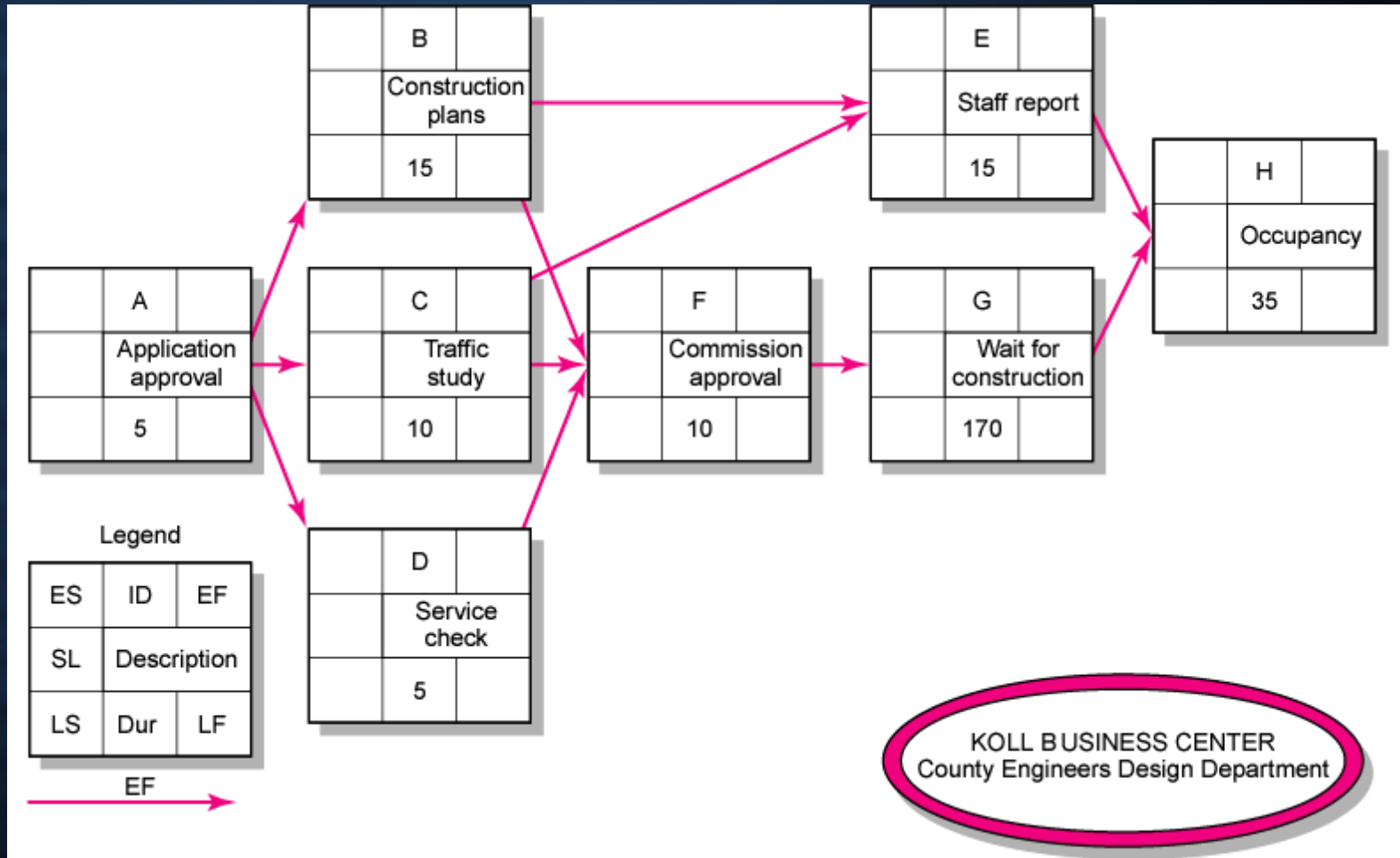


FIGURE 6.5

Activity-on-Node Network Forward Pass

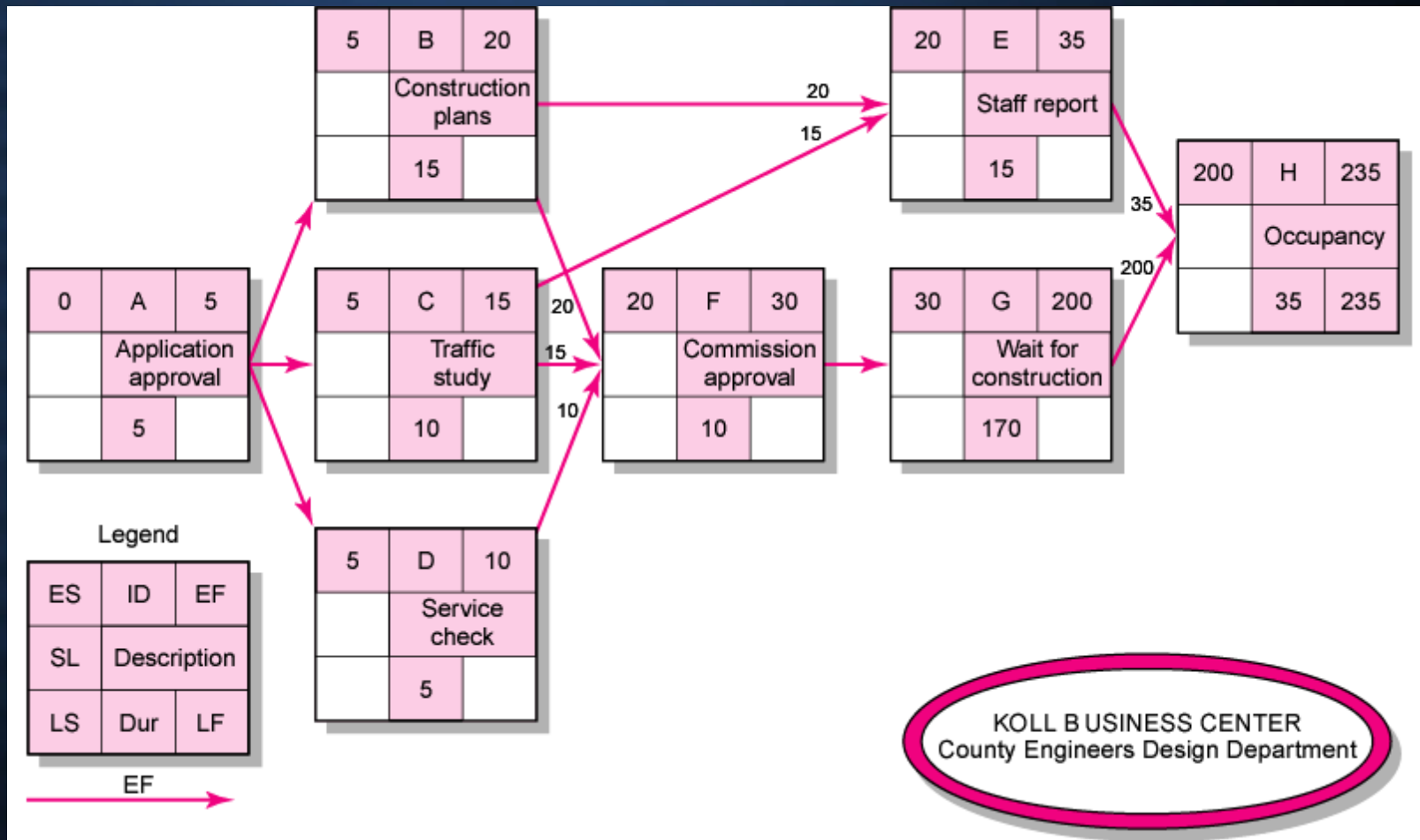


FIGURE 6.6

Forward Pass Computation

- Add activity times along each path in the network ($ES + \text{Duration} = EF$).
- Carry the early finish (EF) to the next activity where it becomes its early start (ES) *unless...*
- The next succeeding activity is a merge activity, in which case the largest EF of all preceding activities is selected.

Activity-on-Arrow Network Backward Pass

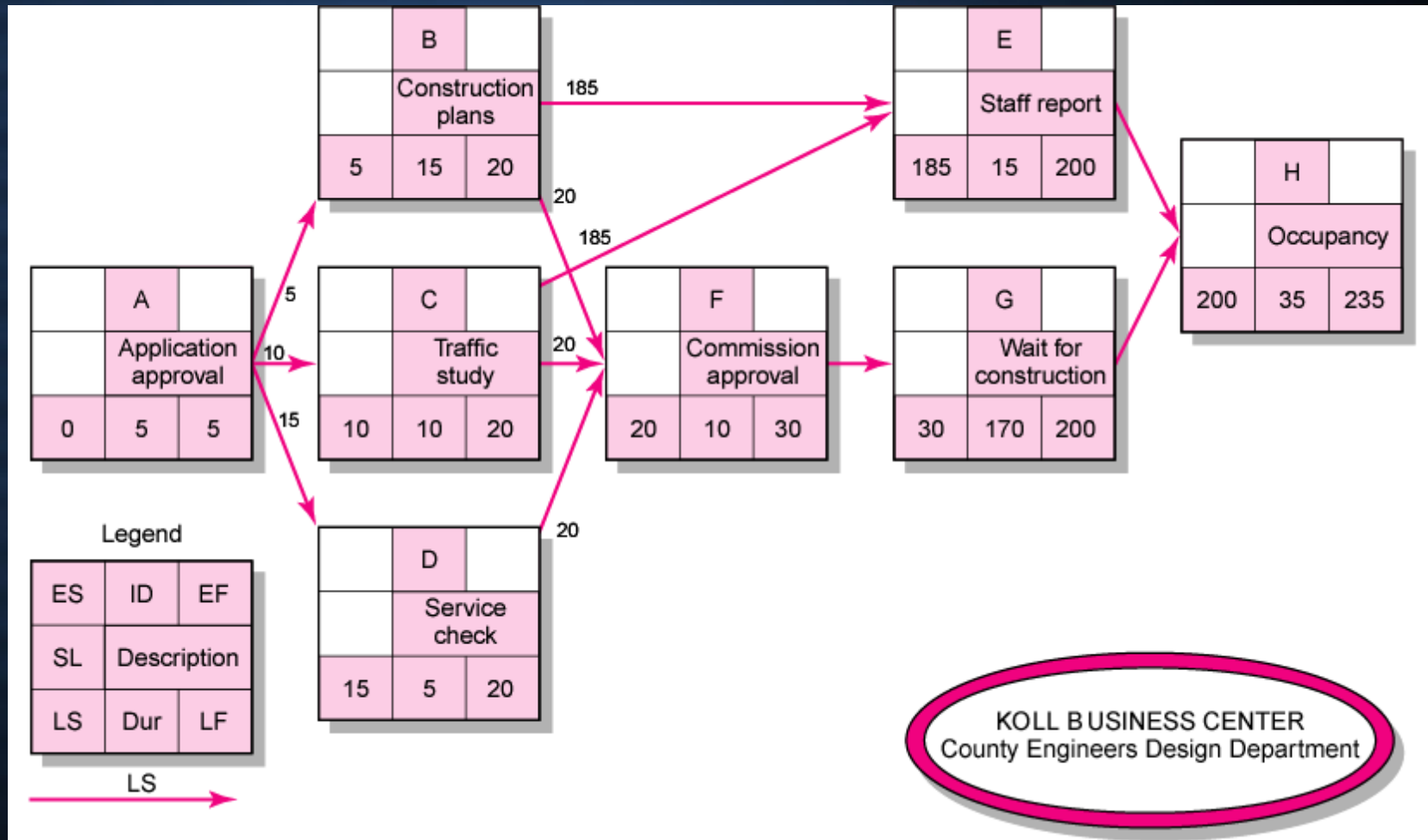


FIGURE 6.7

Backward Pass Computation

- Subtract activity times along each path in the network ($LF - \text{Duration} = LS$).
- Carry the late start (LS) to the next activity where it becomes its late finish (LF) *unless...*
- The next succeeding activity is a burst activity, in which case the smallest LF of all preceding activities is selected.

Determining Slack (or Float)

- Free Slack (or Float)
 - The amount of time an activity can be delayed without delaying connected successor activities
- Total Slack
 - The amount of time an activity can be delayed without delaying the entire project
- The critical path is the network path(s) that has (have) the least slack in common.

Sensitivity of a Network

- The likelihood the original critical path(s) will change once the project is initiated.
 - **Function of:**
 - The number of critical paths
 - The amount of slack across near critical activities

Activity-on-Arrow Network with Slack

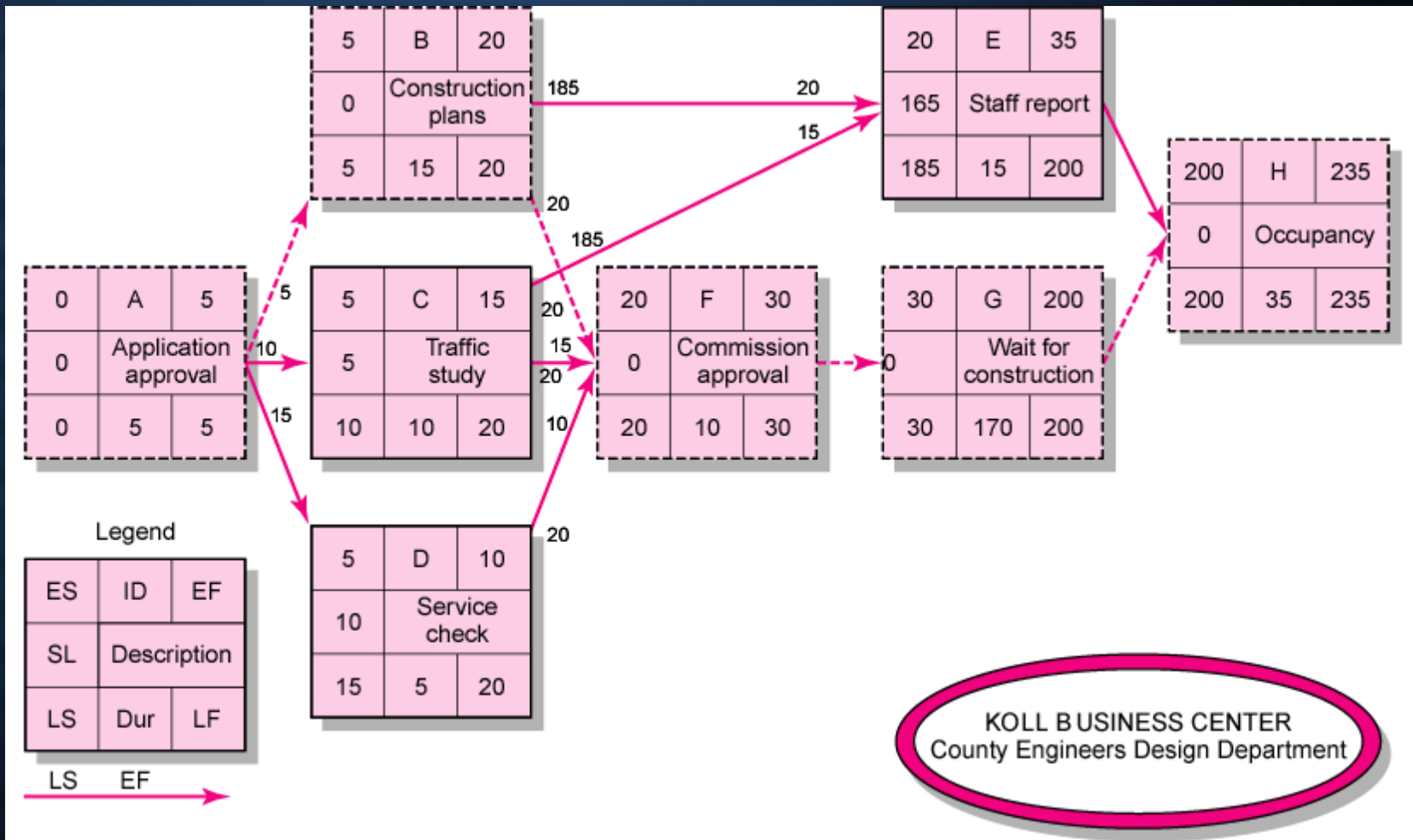


FIGURE 6.8

Practical Considerations

- Network logic errors
- Activity numbering
- Use of computers to develop networks
- Calendar dates
- Multiple starts and multiple projects

Illogical Loop

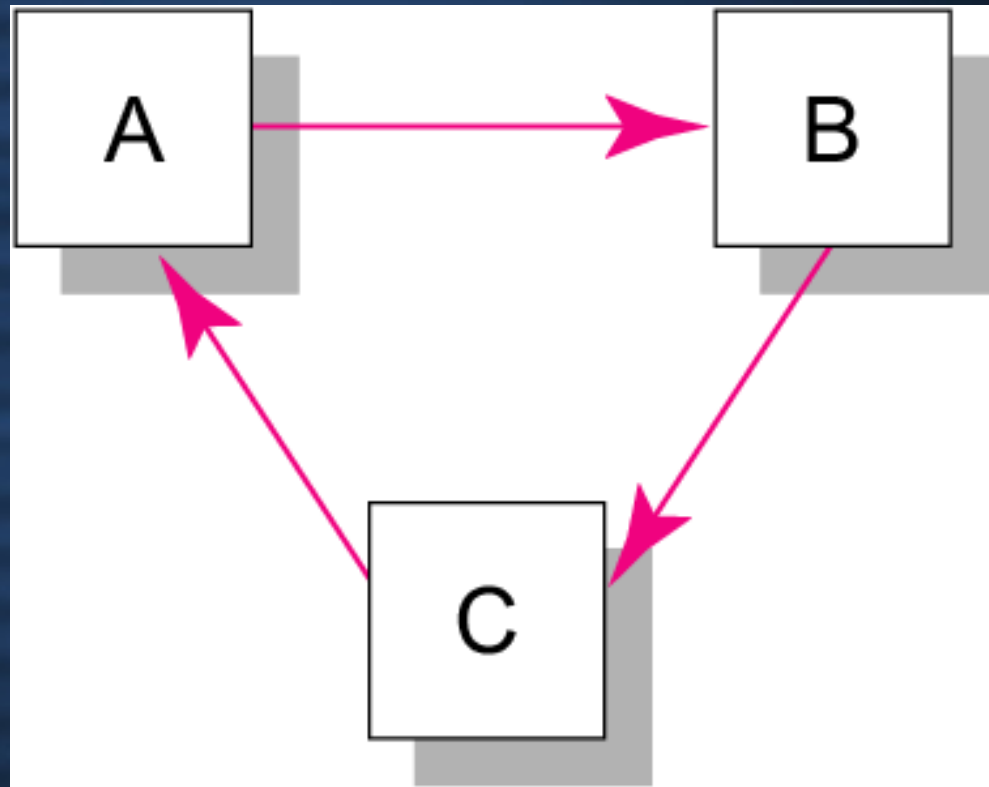


FIGURE 6.9

Air Control Project

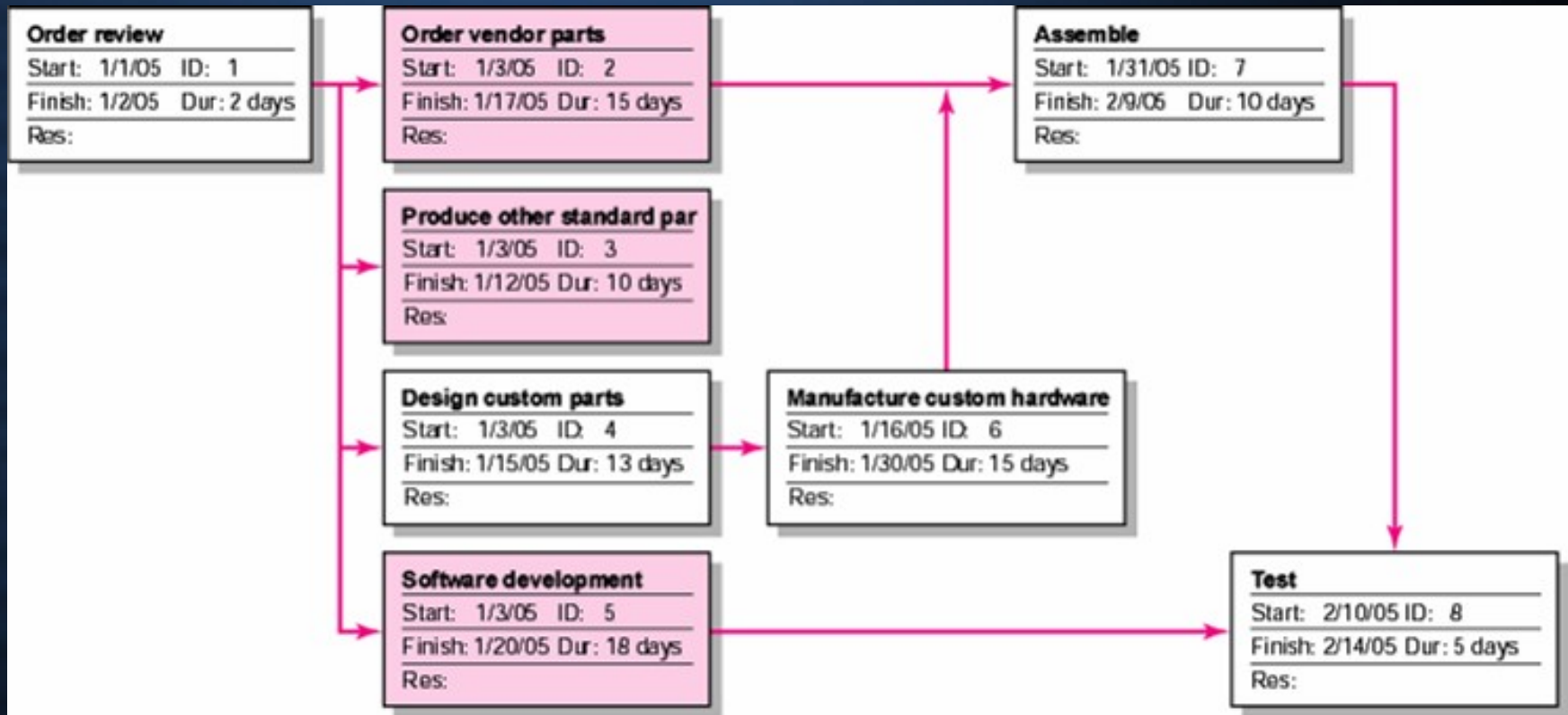


FIGURE 6.10

Air Control Project (cont'd)



FIGURE 6.11

Activity-on-Arrow Network Building Blocks

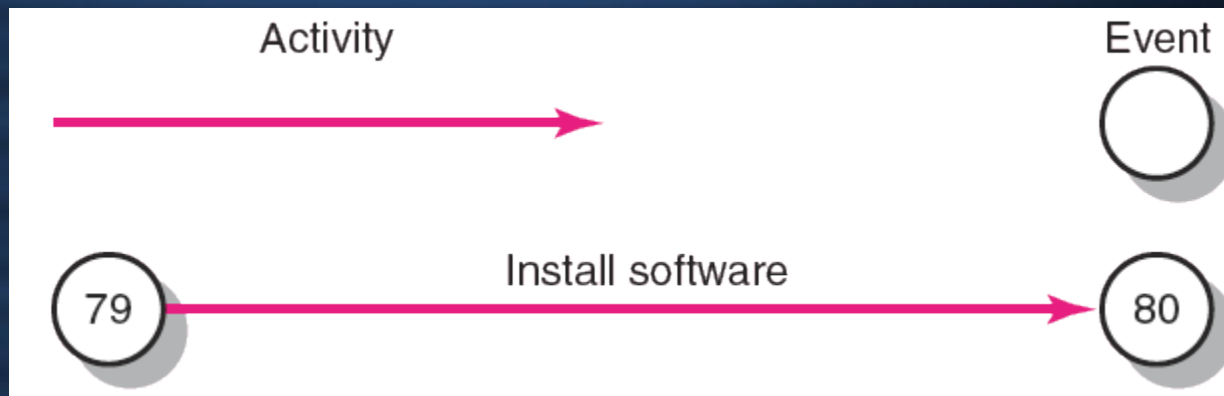


FIGURE A6.1

Activity-on-Arrow Network Fundamentals

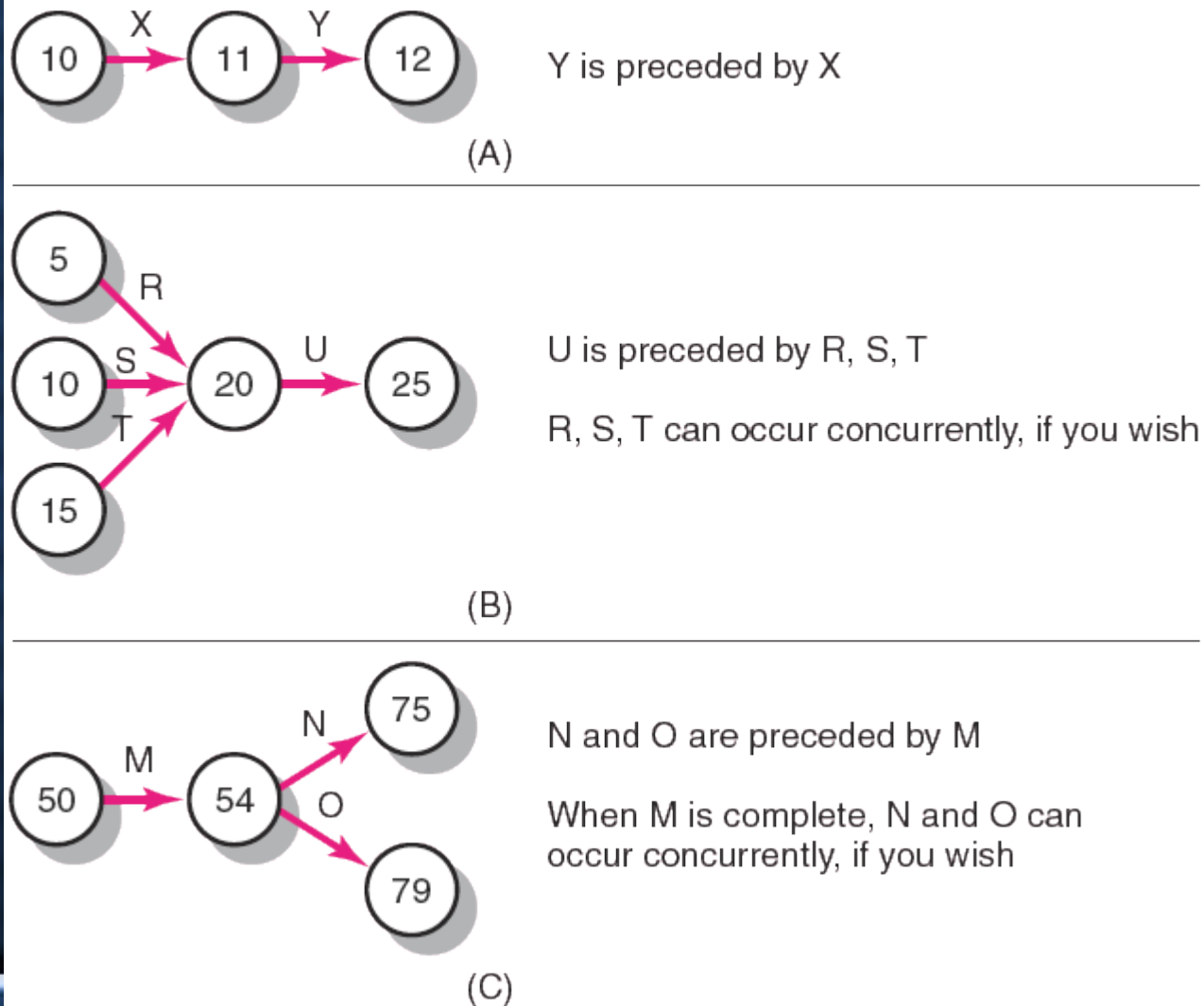
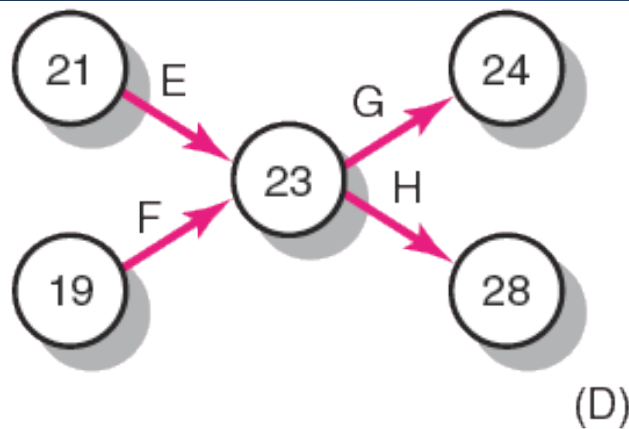


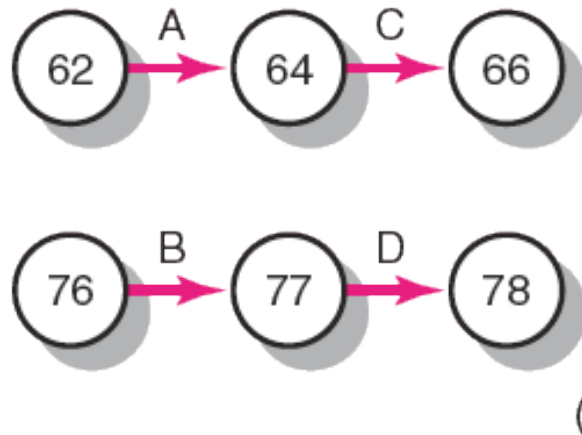
FIGURE A6.2

Activity-on-Arrow Network Fundamentals



E and F must precede G and H

E and F can occur together, if you wish
G and H can occur together, if you wish



A must precede C
B must precede D

Path A–C is independent of path B–D

FIGURE A6.2 (cont'd)

Koll Center Project: Network Information

KOLL BUSINESS CENTER County Engineers Design Department

Activity	Description	Preceding Activity	Activity Time
A	Application approval	None	5
B	Construction plans	A	15
C	Traffic study	A	10
D	Service availability check	A	5
E	Staff report	B, C	15
F	Commission approval	B, C, D	10
G	Wait for construction	F	170
H	Occupancy	E, G	35

TABLE A6.3

Partial Koll Business Center AOA Network

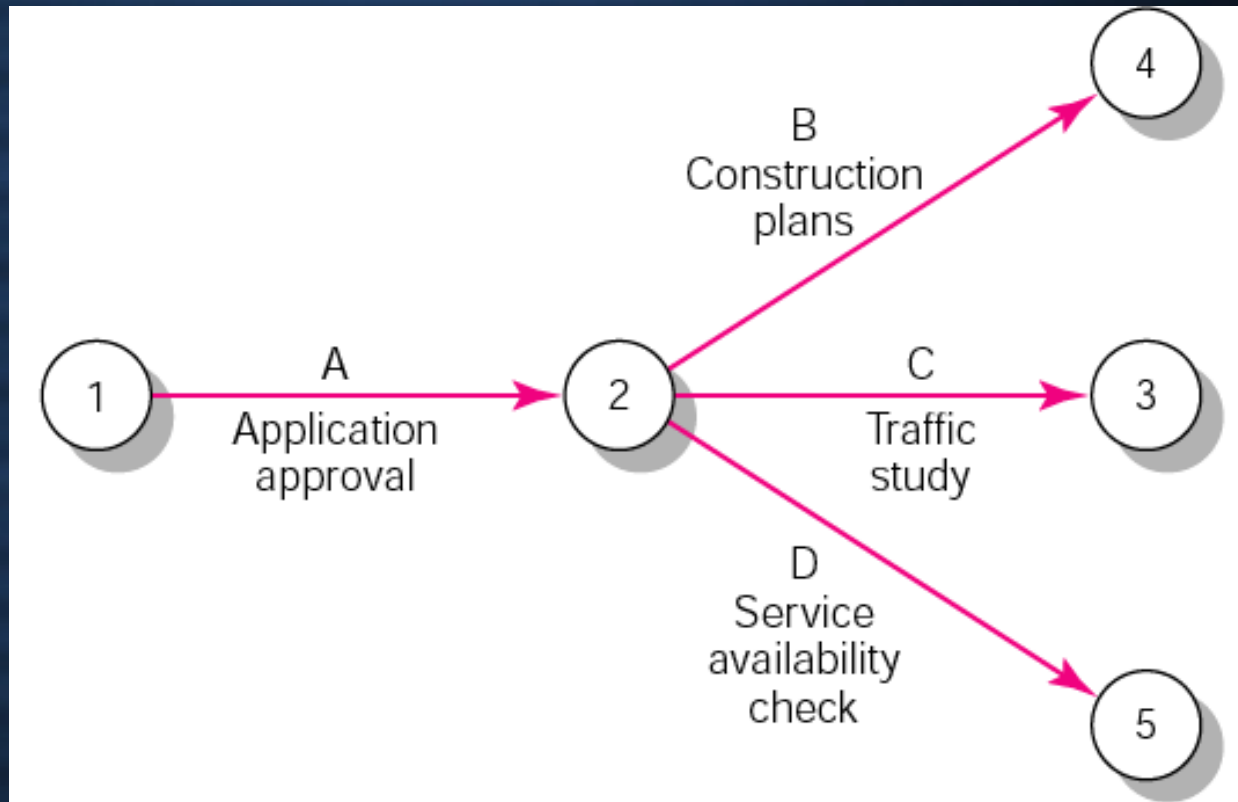


FIGURE A6.3

Partial AOA Koll Network

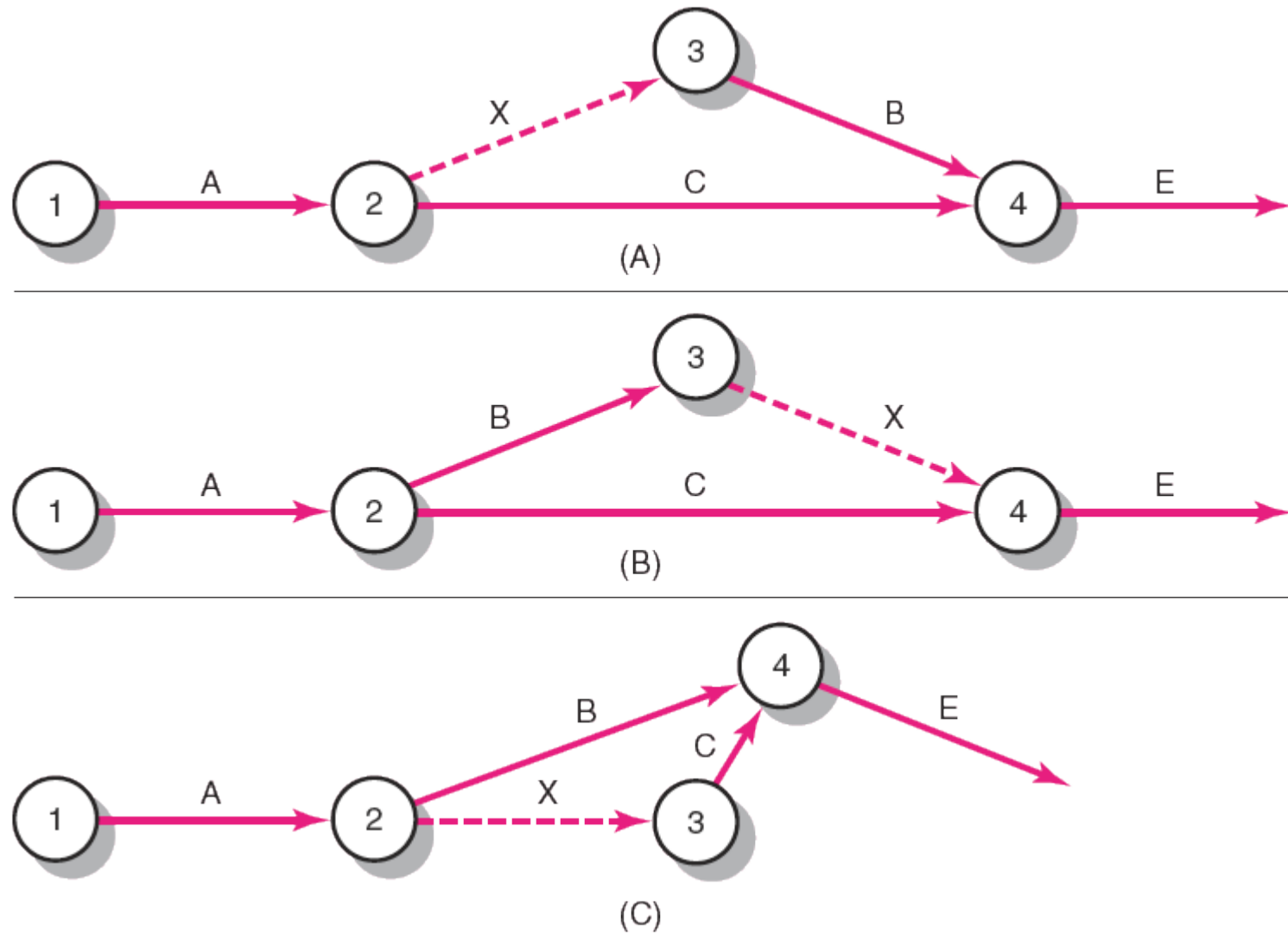


FIGURE A6.4

Partial AOA Koll Network (cont'd)

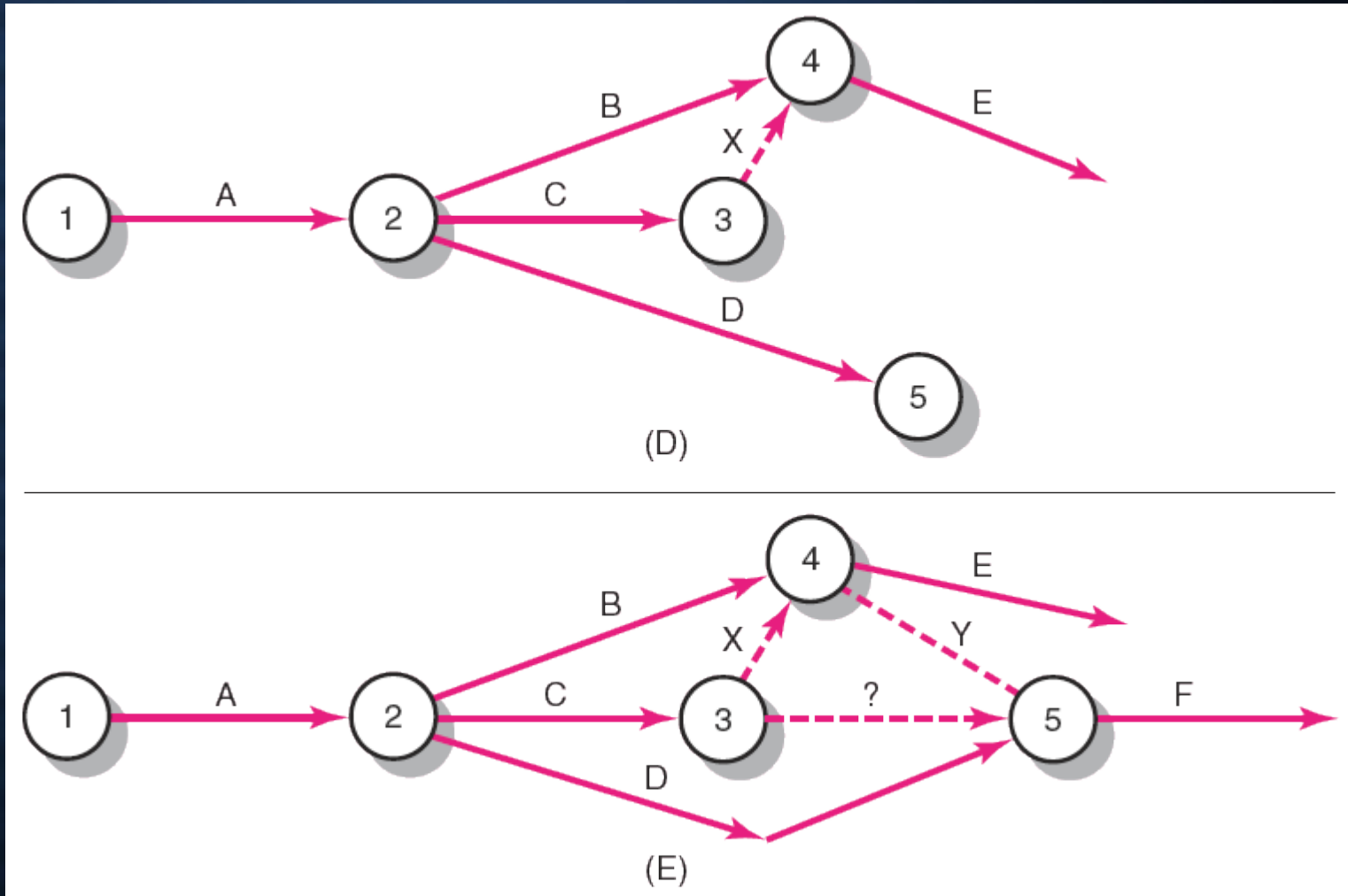


FIGURE A6.4 (cont'd)

Activity-on-Arrow Network

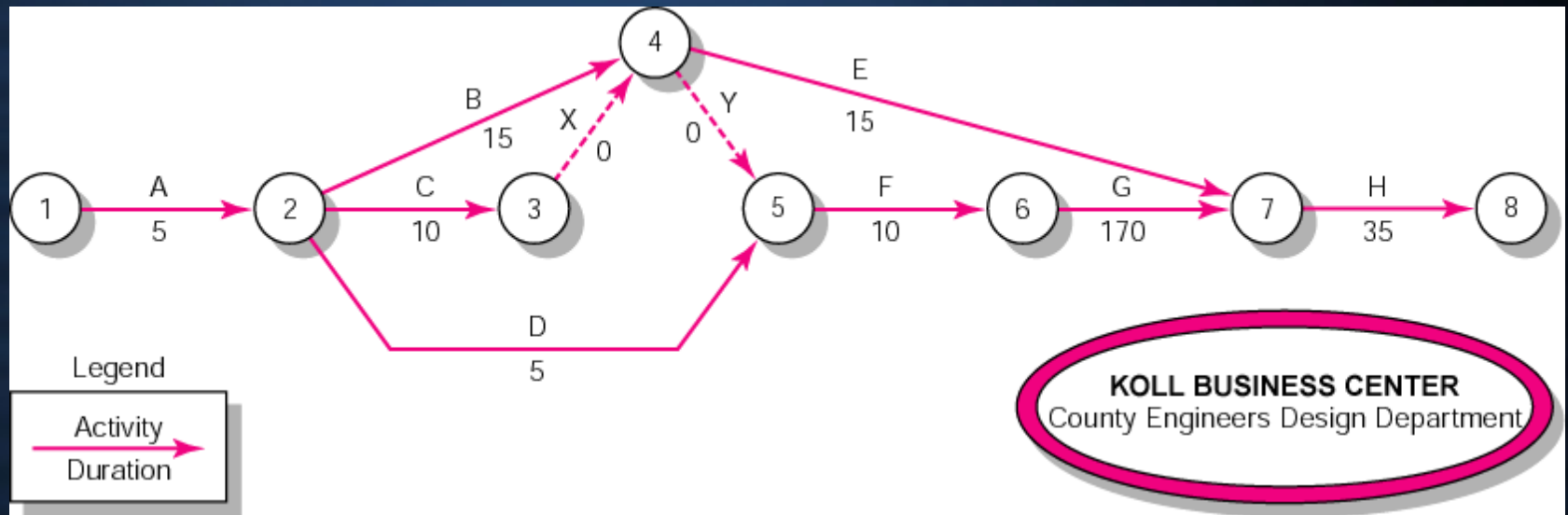


FIGURE A6.5

Activity-on-Arrow Network Forward Pass

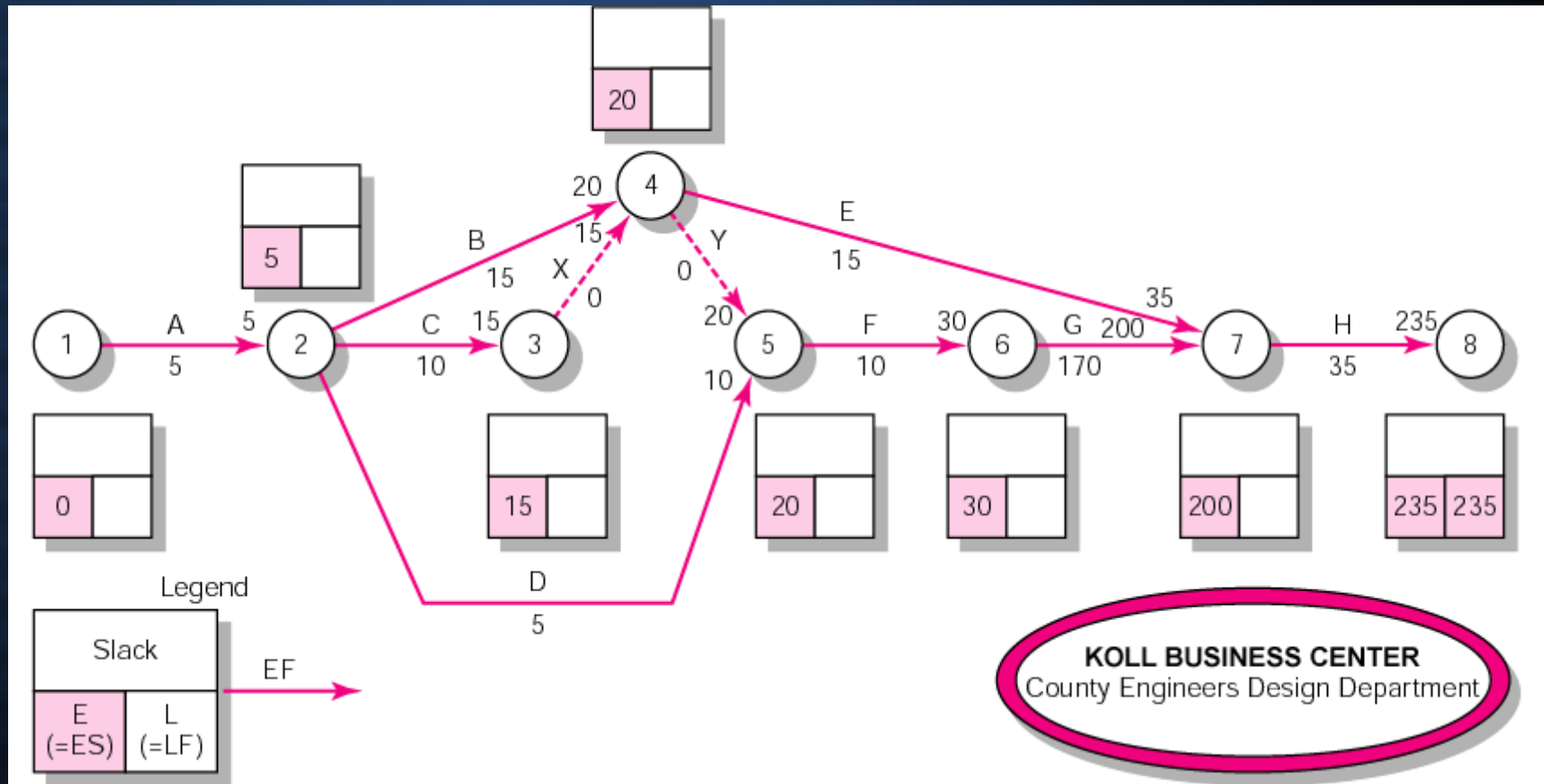


FIGURE A6.6

Activity-on-Arrow Network Backward Pass

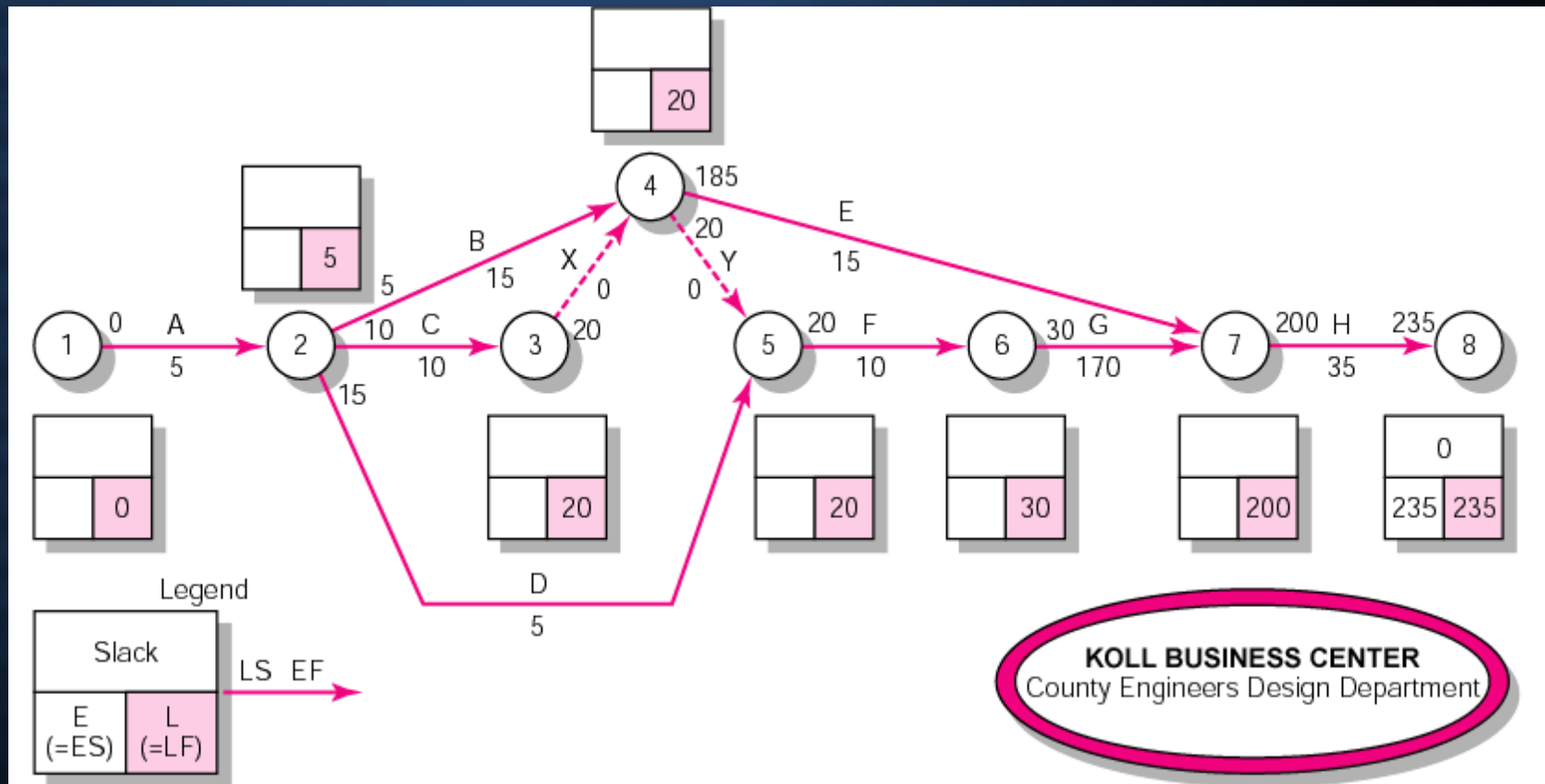


FIGURE A6.7

Activity-on-Arrow Network Backward Pass, Forward Pass, and Slack

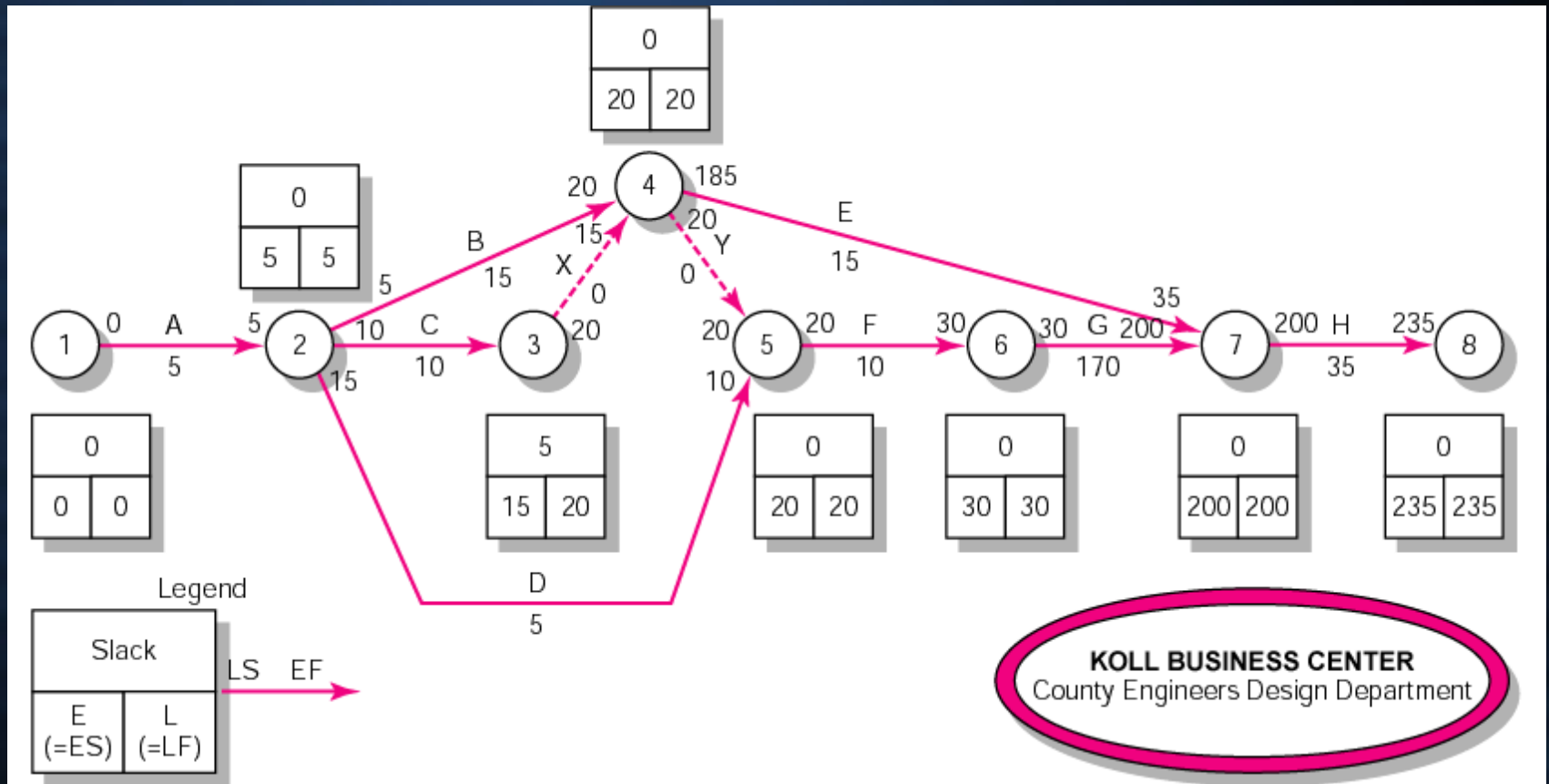


FIGURE A6.8

Air Control Inc. Custom Order Project—AOA Network Diagram

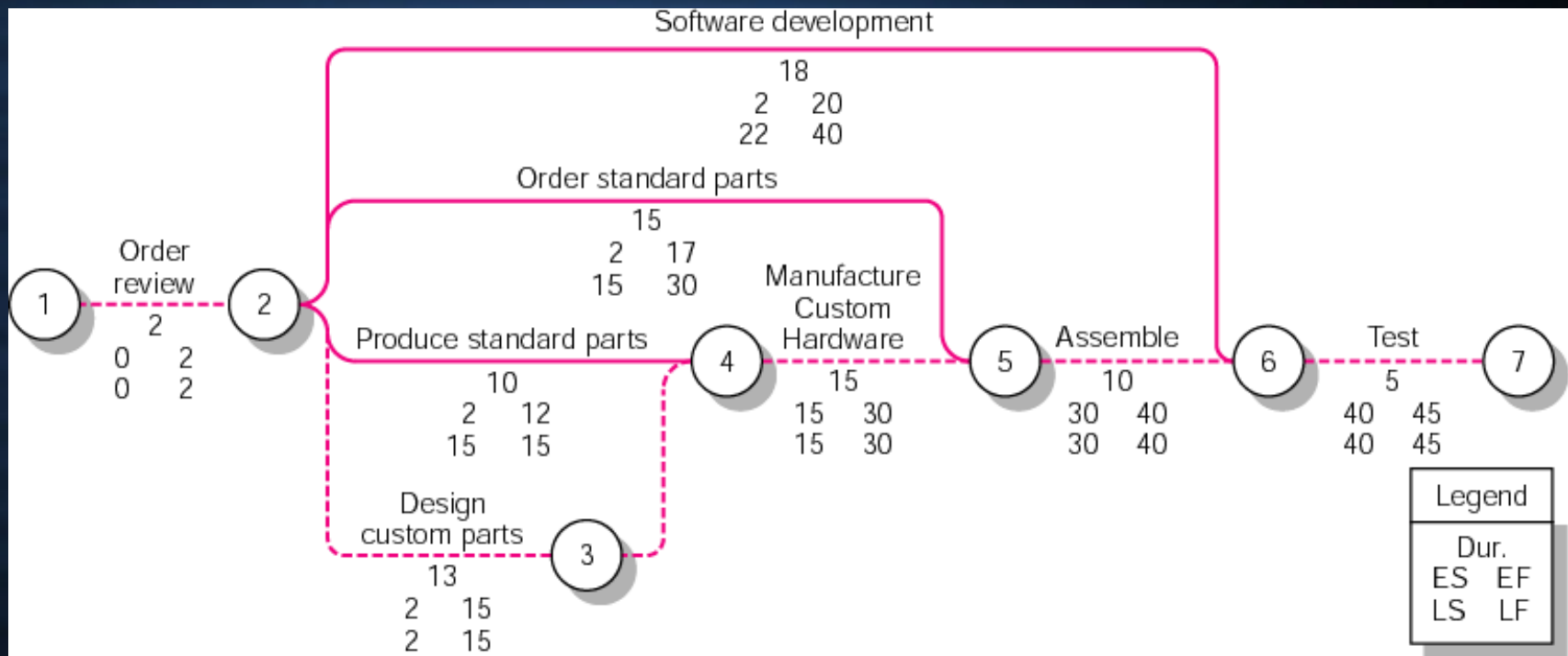


FIGURE A6.9

Comparison of AON and AOA Methods

AON Method

Advantages

1. No dummy activities are used.
2. Events are not used.
3. AON is easy to draw if dependencies are not intense.
4. Activity emphasis is easily understood by first-level managers.
5. The CPM approach uses deterministic times to construct networks.

Disadvantages

1. Path tracing by activity number is difficult. If the network is not available, computer outputs must list the predecessor and successor activities for each activity.
2. Network drawing and understanding are more difficult when dependencies are numerous.

AOA Method

Advantages

1. Path tracing is simplified by activity/event numbering scheme.
2. AOA is easier to draw if dependencies are intense.
3. Key events or milestones can easily be flagged.

Disadvantages

1. Use of dummy activities increases data requirements.
2. Emphasis on events can detract from activities. Activity delays cause events and projects to be late.